Growing up in a compound for doctors due to my dad's job, and with friends studying medicine, I developed an interest in healthcare early on. The first time I worked on the healthtech intersection was after studying ML in my bachelor's. My mind went to how biases in healthcare literature would be cemented in the upcoming age of AI healthcare. I compiled my findings into a blog and messaged researchers working on such models at the time, proposing methods to check fairness. Received a few positive responses, but a lack of resources prevented any actual work. That's what motivated me to pursue a master's and be part of an institute again.

Satisfied with my ML knowledge base, I focused on applying it. I took a health innovations course covering the EU healthtech landscape and the end-to-end process of bringing innovations through it. A hospital project course investigating EHR interoperability challenges in Sweden, learning the gap between implemented and existing standards. A small consultancy project forecasting patient arrivals at an emergency department. Most recently, I've started a pilot with a hospital in Saudi Arabia on using AI to read electrophoresis results and speed up lab turnaround times. None are deep research projects individually, but the breadth of exposure means I understand the structures and common problems. A PhD coming from that awareness is more likely to produce something workers actually want to use, and that fits the system.

The technical work I'm most proud of in my master's is the multimodal pipeline I built for my thesis. I was classifying harmful comments on short-form video platforms, and a portion of the data was image and GIF-based. The pipeline ran a vision-language model to generate descriptions, then fed those into the text classifier. Because I was doing everything alone, from collecting, annotating, to building, I had a much better feel for what would work. I essentially thought, “What information helped me decide if something was harmful?”. So I came up with annotater notes, describing GIFs the same way I describe Gen Z slang to my parents, and it worked great for the accuracy. Similarly, I’d come across comments that, just by their content, could be harmless or harmful. But given the vibe of the entire comment section, it became clear what their intention is, so I included multiple passes in one of the architectures. I finally felt confident in making design choices and knew what was possible and how to implement it, compared to just running the latest models on the data.

The specific problem here, detecting rare endosomal escape events in 4D live-cell data, is interesting from a detection standpoint. Rare event detection in dense, high-dimensional sequential data is hard regardless of the domain. I’m not sure what the interactions between standard vision models and 4D data look like and how we’d handle the inevitable case of high recall and low precision. I’m excited for the chance to develop methods that contribute to unravelling RNA. I’m not a biology student, but I understand RNA could be one of the best programmable vehicles in modern medicine, if only we could comprehend and replicate endosomal escape. My grandmother passed away from lung cancer, and if we could treat patients without the pain of chemo, it would be an amazing step forward.

I don't have a background in microscopy or RNA biology and will need to build that up. The computational foundation is there. I can start in October 2026 or earlier if needed. The earliest realistically is August 2026.